Chapter 9 Study Sheet

Right Triangles and Trigonometry — everything you need on one page.

9.1 The Pythagorean Theorem

- $a^2 + b^2 = c^2$, where c is the HYPOTENUSE. Given the hypotenuse and a leg, subtract.
- Pythagorean triples: 3-4-5 · 5-12-13 · 8-15-17 · 7-24-25 · 20-21-29 (and all their multiples).
- Classify with the largest side c: $c^2 = a^2 + b^2$ right · $c^2 > a^2 + b^2$ obtuse · $c^2 < a^2 + b^2$ acute. Check the Triangle Inequality first.
- Simplest radical form: pull out perfect squares ($\sqrt{80} = 4\sqrt{5}$); no radicals in denominators.

9.2 Special Right Triangles

- 45°-45°-90°: hypotenuse = leg · $\sqrt{2}$. From the hypotenuse, divide by $\sqrt{2}$ (= multiply by $\sqrt{2}/2$).
- 30°-60°-90°: hypotenuse = 2 · shorter leg; longer leg = shorter leg · $\sqrt{3}$. The shorter leg is opposite 30°.
- Squares hide 45° triangles (diagonal); equilateral triangles hide 30°-60°-90° (altitude).

9.4–9.5 Trigonometric Ratios

- SOH-CAH-TOA: sin = opp/hyp · cos = adj/hyp · tan = opp/adj — always labeled FROM the angle in use.
- To find a side: pick the ratio linking the side you know with the side you want; cross-multiply.
- If x is in the DENOMINATOR (x = known/ratio), divide instead of multiplying.
- Cofunction fact: sin A = cos B when A + B = 90°.
- Elevation/depression: horizontal line first; angle at the observer; nearly always a tangent equation; the depression from A to B equals the elevation from B to A.

9.6 Solving Right Triangles

- 'Solve' = find all three sides and all three angles.
- Angles from sides: $\sin^{-1}$, $\cos^{-1}$, $\tan^{-1}$ take a ratio and return the angle.
- Standard plan: third side by Pythagorean Theorem → one acute angle by an inverse ratio → other angle = 90° − first.
- Minimum information: two sides, or one side + one acute angle.